CP–7807

Sub. Code
11

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

ENGLISH

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What is the imaginary elixir of life?
2. Give an account of Mrs. Packletide's Tiger?
3. What is the theme of Ozymandias?
4. Who is Ulysses?
5. What is waterloo?
6. What happened on Mother's Day?
7. How do you develop comprehension skills?
8. Give some tips to develop communication skills.

Part B (5 × 12 = 60)

Answer **all** the questions.

9. (a) How does C.V. Raman prove that water is the real elixir of life?

Or

(b) Discuss Dicken's impressions of the journey to Niagara.

10. (a) How does Goldsmith portray the village School Master?

Or

(b) How does naturalism express in "The Solitary Reaper"?

11. (a) Explain the adventurous spirit revealed in "Ulysses".

Or

(b) Discuss the main idea of "The Proposal".

12. (a) Develop the hints into an essay :

Two friends – went to forest – suddenly a bear – came near – one boy climbed the tree quickly – escape from the bear – another boy – helpless – lay down on the ground silently – bear came close – watched – went away – the boy on the tree came down – asked him – what did bear say – in his ears – boy replied – don't believe friends who leave you when in trouble.

Or

(b) Write a letter to your brother informing him for your success in the NET exam.

13. (a) Discuss the need for effective communication skills in the present scenario.

Or

(b) Prepare a welcome address for sports day in your college.

CP–7808

Sub. Code
12

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

MATHEMATICS

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Resolve into partial fractions : $\dfrac{5x+1}{(x-1)(x+2)}$.

2. Find the coefficient of x^n in the expansion of $(1+2x+3x^2+4x^3+\cdots \text{to } \infty)^2$.

3. Find the real and imaginary part of $\dfrac{(\cos 3\theta + i\sin 3\theta)^{-2}\,(\cos 2\theta - i\sin 2\theta)^{3/2}}{(\cos 5\theta - i\sin 5\theta)^3\,(\cos 2\theta + i\sin 2\theta)^7}$.

4. Express $\tan 6\theta$ in terms of $\tan\theta$.

5. Find the nth derivative of $y = e^{2x}\sin^2 x$.

6. State Leibnitz theorem.

7. Evaluate $\int_0^{\pi/2} \cos^2 x\, dx$.

8. Solve $(D^2+9)y = 0$.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Show that, when n is a positive integer,

$$1-\frac{n+x}{1+x}+\frac{(n+2x)(n-1)}{2!(1+x)^2}-\frac{(n+3x)(n-1)(n-2)}{3!(1+x)^3}+\cdots=0.$$

Or

(b) Expand $(1+x+x^2+x^3)^{1/3}$ upto the terms in x^4.

10. (a) Express $\cos 6\theta$ and $\frac{\sin 6\theta}{\sin\theta}$ in series of powers of $\cos\theta$.

Or

(b) Using De Moivre's theorem expand $\cos^8\theta$ in a series of cosines of multiples of θ.

11. (a) If $y=(x+\sqrt{1+x^2})^m$, prove that $(1+x^2)y_{n+2}+(2n+1)xy_{n+1}+(n^2-m^2)y_n=0$.

Or

(b) If $u=\tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\sin 2u$.

12. (a) Evaluate

(i) $\int \sec x \, dx$

(ii) $\int \frac{dx}{1+\tan x}$.

Or

(b) Evaluate $\int \frac{3x+1}{(x-1)^2 (x+3)} dx$.

13. (a) Solve :

(i) $xp^2 - 2yp + x = 0$

(ii) $xyp^2 + p(3x^2 - 2y^2) - 6xy = 0$.

Or

(b) Solve :

(i) $y = (x-a)p - p^2$

(ii) $y = xp + x(1+p^2)^{1/2}$.

CP–7809

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13

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

BASIC ELECTRICITY

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define Coulomb and Ampere.
2. Explain about paramagnetic materials.
3. State and explain Ohm's law.
4. Write a short note on Switch and its uses.
5. Briefly explain the working of voltmeter.
6. Explain types of transformers.
7. What is Q factor of a coil?
8. Explain Wattless current.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Discuss in detail the types of magnetic materials and their applications.

Or

(b) Define mutual induction and derive an expression for mutual induction between two coils.

10. (a) State and explain Kirchoff's law with an example.

Or

(b) Discuss in detail :

(i) Transformer

(ii) Voltage regulator.

11. (a) Describe the construction and working of multimeter with a neat diagram.

Or

(b) Discuss in detail about DC motor.

12. (a) Discuss in detail about generation of Alternating voltages and current and its RMs value.

Or

(b) Obtain an expression for current in an AC circuit containing resistance and inductance as well as resistance capacitor.

13. (a) Deduce the resonance relation in the series LCR circuits.

Or

(b) (i) Obtain the expression for power in an AC circuit containing only resistance and inductance.

(ii) Two coils of inductances 0.5 henry connected in parallel with one coil of 0.75 h in series. Determine the equivalent inductance of the circuit.

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CP–7810

Sub. Code
21

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

PRINCIPLE OF ELECTRONICS AND ELECTRONIC CIRCUITS

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define capacitance. Give its unit.

2. What do you mean by Q-factor of a coil?

3. What is meant by intrinsic semiconductor? Give examples.

4. Give three pentavalent impurities.

5. Define pinch off voltage in FET.

6. Mention some uses of UJT.

7. What is a class B amplifier?

8. State Barkhausen criteria for oscillation.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Write a note on :

(i) Fixed resistor

(ii) Variable resistor

(iii) Colour coding.

Or

(b) Find the effective value of two inductances connected in

(i) Series

(ii) Parallel.

10. (a) Describe the working of pn junction diode under forward and reverse bias. Hence discuss its V-I characteristics.

Or

(b) Describe the method of studying the characteristics of an NPN transistor under CE configuration.

11. (a) Explain the construction and V-I characteristics of FET.

Or

(b) Describe the construction and V-I characteristics of DIAC.

12. (a) With a neat circuit diagram, explain the operation of class A-amplifiers.

Or

(b) Draw and explain the operation of complementary pair amplifier.

13. (a) Draw the circuit diagram of a Hartley oscillator and explain its working.

Or

(b) With a neat circuit diagram, explain the working of Wein bridge oscillator.

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CP–7811

Sub. Code
22

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

APPLIED PHYSICS

(Upto 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A ($5 \times 3 = 15$)

Answer any **five** questions.

1. Define interface in plane parallel thin film.

2. Define resolving power of telescope.

3. Define Twin – Paradox concept.

4. Define Ultrasonics in sound waves.

5. Define population inversion.

6. Define holography.

7. Define super conductivity.

8. Define Meissner effect.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain the formation of Newton's rings by interference.

Or

(b) Explain diffraction grating.

10. (a) Explain the concept of polarization by Nicol Prism.

Or

(b) Explain Gallilieon transformation in relativity concept.

11. (a) Explain acoustic materials used in Aviation.

Or

(b) Explain various methods of Non–Destructive Testing.(NDT)

12. (a) Explain the concept of Semi conductor laser.

Or

(b) Explain the types of optical fibres.

13. (a) Explain Bravass lattices.

Or

(b) Explain Miller's Indices.

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CP–7812

Sub. Code
23

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Aircraft Maintenance Science

STRENGTH OF MATERIALS AND APPLIED MECHANICS

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define Statics and Dynamics.

2. Define :

   (a) Factor of safety

   (b) Poisson's Ratio.

3. State Parallelogram Law of Forces

4. Define Hooke's Law of Elasticity

5. What are the different types of gears used in aviation maintenance?

6. Define Moment of force

7. Mention various types of loads and beams with diagram.

8. Define Stress and Strain.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain fundamental and derived units with examples.

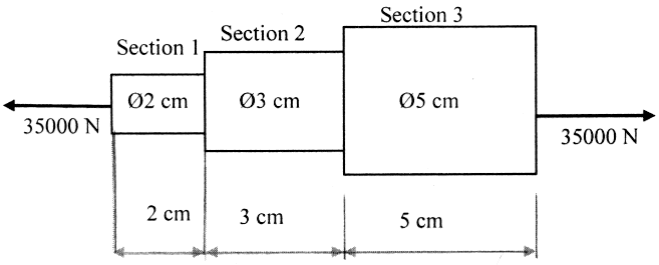
Or

(b) Explain various types of forces in a system.

10. (a) An axial pull of 35000 N is acting in a bar consisting of three lengths as shown in the fig. If the Young's Modulus is 2.1×10^5 N/mm^2, determine

(i) Stress in each section

(ii) Total extension of the bar.



Or

(b) Explain Stress-Strain Relationship of materials.

11. (a) Explain Shear Force diagrams and Bending Moment diagrams.

Or

(b) Explain different types of Trusses, with neat sketches.

12. (a) What are the various types of rivets? Explain the procedure for riveting.

Or

(b) Explain the various materials used for bearings and springs in aviation maintenance.

13. (a) Explain various types of Belt Drives with neat sketches.

Or

(b) Briefly explain :

(i) Parallelogram Law of Forces

(ii) Polygon Law of Forces.

CP–7813

Sub. Code
24

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Aircraft Maintenance Science

FUNDAMENTALS OF THERMODYNAMICS

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What is a PMM1? Why is it impossible?
2. Define zeroth law and first law of thermodynamics.
3. Define entropy.
4. Define the terms source and sink.
5. What is mach number
6. What is a black body?
7. Explain the mixed or dual cycle.
8. What is refrigerating effect?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Differentiate between point function and path function.

Or

(b) Explain the specific heats at constant volume and constant pressure.

10. (a) Differentiate between heat pump and refrigerator.

Or

(b) Briefly discuss about the concept of entropy.

11. (a) What is the impulse function? What is a fanno line?

Or

(b) Explain the occurence of choking for adiabatic flow with function and for diabetic flow without friction.

12. (a) What do you understand by natural convection and forced convection?

Or

(b) Why are counter flow heat exchangers superior to parallel flow heat exchangers?

13. (a) Mention the merits and demerits of the stirling and Ericsson cycles.

Or

(b) Explain with a line diagram the construction and working of a vapour compression system of refrigeration.

CP–7814

Sub. Code
31

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

MECHANICS OF FLIGHT AND AIRCRAFT PERFORMANCE

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Briefly explain the terms density and pressure.

2. What are the various factors which contribute to the drag produced by an aircraft?

3. Describe angle of attack.

4. Write the fundamental equation of drag and its abbreviations.

5. Define — Aspect ratio.

6. How is an Airplane's load factor determined?

7. Describe the three axes of an airplane.

8. Define – Take off.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) (i) Explain — Bernoulli's principle. (4)

(ii) Discuss how lift generates in an aircraft by using Newton's law of motion. (8)

Or

(b) Explain the following: (3 marks each)

(i) Aerodynamics.

(ii) Inversion.

(iii) Stagnation pressure

(iv) Downwash angle.

10. (a) Explain — Airfoil Terminology

Or

(b) Discuss:

(i) Reynolds' Number. (4 marks each)

(ii) Center of Pressure.

(iii) Angle of attack Vs Lift

11. (a) Explain in detail about the characteristics of an airfoil.

Or

(b) Explain the following:

(i) Angle of Incidence (6 marks each)

(ii) Effects of Flaps.

12. (a) Write the function of:

(i) Rudder with neat sketch (6 marks each)

(ii) Aileron

Or

(b) Explain the lateral stability of an aircraft in detail.

13. (a) What are the factors affecting take off and climb? Explain in details.

Or

(b) Discuss the effect of changes of engine power.

CP–7815

Sub. Code
32

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

FLUID MECHANICS

(Upto 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define compressibility and incompressibility.

2. What are the types of mechanical gauges?

3. Write short notes on discharge.

4. What are the losses in pipe?

5. Write the expression for power transmission through pipe.

6. What are the types of turbine?

7. Short notes on (a) accumulator (b) intensifier.

8. Define centrifugal pumps.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain in detail about measurement of pressure.

Or

(b) Short notes on :

(i) Density

(ii) Capillary

(iii) Viscosity

(iv) Surface tension

(v) Specific gravity

(vi) Specific weight.

10. (a) Explain in detail about types of fluid flow.

Or

(b) Derive Euler's Bernoulli's equation.

11. (a) Write brief notes on losses in flow through pipe.

Or

(b) Losses in Pipe (with neat sketch)

(i) Parallel pipe

(ii) Branched pipe.

12. (a) Force exerted by the jet on hinged plate.

Or

(b) Explain any four types of turbine with sketch.

13. (a) (i) Hydraulic press

(ii) Accumulator.

Or

(b) (i) Hydraulic intensifies.

(ii) Hydraulic lift.

CP–7816

Sub. Code
33

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

AIRCRAFT MATERIALS AND NON DESTRUCTIVE TEST

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A $(5 \times 3 = 15)$

Answer any **five** questions.

1. Define malleability.
2. Differentiate ferrous and non ferrous metals.
3. Explain the properties of C_4.
4. Define spheroidising.
5. Describe the construction of plywood.
6. Define composite material.
7. What is the importance of NDI?
8. Write short notes on Inconel.

Part B $(5 \times 12 = 60)$

Answer **all** questions.

9. (a) Explain :

   Hardness, Brittleness, Tensile strength, Ductility, Malleability.

   Or

   (b) Explain :

   Tensile test and Izod impact test.

10. (a) Explain in detail about chromium and its alloys.

Or

(b) Explain the properties of titanium and its alloys.

11. (a) Explain the advantages of Hardening and tempering.

Or

(b) Explain the process of heat treatment for non ferrous metals and its alloys.

12. (a) Explain about Reinforced materials used in aircraft construction.

Or

(b) (i) Explain various types of studs, screws, washers, used in aircraft construction.

(ii) Explain Riveting procedure.

13. (a) Explain Dye penetrant inspection.

Or

(b) Explain Radiography and Magnaflux inspection.

CP–7817

Sub. Code
34

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

AIRFRAME STRUCTURE AND SYSTEMS

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Explain the various structural stresses acting on the aircraft.
2. What is the purpose of spars and ribs in wing construction? Explain.
3. What is the purpose of trunnion in Landing gear?
4. What is flap? Name its type.
5. Explain the zoning of aircraft.
6. Define ballast and its type.
7. Where the flexible hoses are used?
8. What is the purpose of cross feed fuel system?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Briefly explain the operation of nose wheel steering system.

Or

(b) What are the types of fuselage construction and explain the semi-monocoque construction?

10. (a) Explain conventional flying Control system components and its functions.

Or

(b) Explain the control around vertical axis.

11. (a) How incidence angle of a wing can be checked during rigging?

Or

(b) What are the occasions for checking the structural alignment? Describe the symmetry Checks in detail.

12. (a) With neat sketch explain the types of hydraulic accumulators.

Or

(b) State the advantages and disadvantages of pneumatic system over hydraulic system.

13. (a) Write the general rules to be followed by the aircraft operator during defueling of an aircraft.

Or

(b) Explain the aircraft fuel tanks.

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CP–7818

Sub. Code
41

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

PISTON ENGINE AND PROPELLER

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Differentiate Valve lead and valve lag.
2. Define firing order and write the sequence of 12 cylinder of radial engine.
3. What do you understand by heat range of spark plug?
4. Write the requirements of lubricating oil used in engine lubrication system?
5. Write the basic characteristics of fuel used in aviation.
6. What is Ignition Harness?
7. What is direct type starter?
8. Write the importance of wooden propeller.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Describe four stroke fire events of petrol engine cycle with P.V diagram?

Or

(b) Write short notes on:

(i) Brake Horse Power

(ii) Indicated Horse Power

(iii) Thermal Efficiency

(iv) Crank shaft.

10. (a) Write down the constructional details of various types of pistons.

Or

(b) Draw an external type supercharger system and explain.

11. (a) Describe float type carburetor and its components with diagram.

Or

(b) Explain the operation of dry sump lubrication system of reciprocating engine.

12. (a) Explain the construction of a spark plug in detail.

Or

(b) Explain about the engine Starter motor.

13. (a) Explain in detail about various types of propellers.

Or

(b) Describe the fixed and variable pitch propeller.

CP–7819

Sub. Code
42

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

TURBINE ENGINE

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write the application of Newton law of motion in gas turbine engine.

2. What is the effect of altitude and temperature in the effect of thrust production?

3. What are the functions of inlets?

4. Write the types of turbines used in gas turbine engine.

5. List out the types of fuel tanks and its significance.

6. What is the significance of air turbine starter?

7. Define glow plug.

8. Write the types of propellers used in turbo prop engine.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Write short notes on (i) water/methanol injection (ii) effect of thrust with respect to engine speed and density.

Or

(b) Justify, which type of gas turbine is very suitable for long range and passenger transport aircraft?

10. (a) Write the construction details of (i) thrust reverser (ii) diffuser.

Or

(b) Write the features of noise pollution of gas turbine engine.

11. (a) Describe the Gravity feed fuel system of engine.

Or

(b) Write short notes on (i) Types of fuels (ii) Fuel tanks (iii) Fuel piping system.

12. (a) List out the various types of starters used in gas turbine engine and its features.

Or

(b) List out the types of igniters and its features.

13. (a) List out the advantages and disadvantages of turbo propeller engine.

Or

(b) Describe about the turbine section of turbo propeller aircraft.

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CP–7820

Sub. Code
43

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

INSTRUMENTS AND COMPASS

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define ICAO Q – code of Altimeter.
2. Differentiate extension lead and compensating lead.
3. What are the various methods to measure the atmospheric pressure?
4. What are the errors occurs in magnetic compass?
5. Define pressure altitude and indicated altitudes.
6. Write short notes on magnetic dip and magnetic variation.
7. How EPR measured in Jet engine?
8. Define EGT probe and its material.

Part B (5 × 12 = 60)

Answer **all** questions choosing either (a) or (b).

9. (a) Briefly explain Electronic Display.

Or

(b) Define grouping of aircraft instrument and its layout in cockpit of aircraft.

10. (a) Briefly explain Pitch-Roll erection.

Or

(b) Briefly explain construction and operation of Rate-Gyro-Scope.

11. (a) Briefly explain principle of operation of Stall warning and Altitude alerting system.

Or

(b) What do you mean by Mach number? Briefly explain construction and operation of Mach meter.

12. (a) Briefly explain construction and operation of integrated fuel flow meter.

Or

(b) Briefly describe torque monitoring system of aircraft.

13. (a) Define Coefficient of deviation of magnetic compass and its construction.

Or

(b) Write short notes on :

(i) Terrestrial magnetism

(ii) Hard iron and soft iron magnetism

(iii) Error in compass.

CP–7821

Sub. Code
44

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

AIRCRAFT ELECTRICAL SYSTEM

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (3 × 5 = 15)

Answer any **five** questions.

1. What is invertors?
2. Describe electrolyte.
3. What are the characteristics of electric wire?
4. What is bonding and shielding?
5. Write about thermal runaway.
6. What is purpose of hydrometer test?
7. Describe proximity sensor.
8. What is residual magnetism?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain static discharger with the help of a diagram.

Or

(b) Describe the construction of Nickel Cadmium battery.

10. (a) Draw a diagram of reverse current cutout relay and explain.

Or

(b) Explain the procedure for generator load balancing.

11. (a) Explain parallel electric power distribution system.

Or

(b) Write general requirements for inspection and maintenance of electrical installation.

12. (a) Explain the aircraft fire protection system.

Or

(b) Write short notes on followings :

(i) Instrument lights

(ii) Cabin lights

(iii) Warning lights.

13. (a) Explain the procedure for routing of electric wire bundle.

Or

(b) Discuss about the characteristics of soldered and crimped connections.

CP–7822

Sub. Code
51

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

FUNDAMENTALS OF COMPUTERS AND OFFICE AUTOMATION

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A $(5 \times 3 = 15)$

Answer any **five** questions.

1. Convert $(1001)_2$ to decimal $(\quad)_{10}$.

2. Write the functions of OS.

3. Write steps to italicize a text.

4. What are the indicators that appear in status area?

5. What do you mean by table in database?

6. What is the purpose of Fill Series option?

7. What is the use of AutoSave option?

8. Define the term Query.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Give detailed answer for the following :

   (i) Characteristics of computers

   (ii) Types of software.

Or

   (b) Draw logic circuit diagrams and verify the truth table of AND, OR, NOT, XOR, NAND, NOR gates.

10. (a) Explain any 8 formatting features in Word processing software.

Or

   (b) Explain various methods to cut, copy and paste a paragraph in Word.

11. (a) Discuss various types of charts available in MS Excel.

Or

   (b) Explain the following in spreadsheet.

   (i) Formatting Cell Contents

   (ii) Fixing the size and hiding Rows and columns

   (iii) Autoformatting

   (iv) Adding, deleting and renaming a Sheet

   (v) Conditional formatting

   (vi) Style.

12. (a) How will you create and manipulate slides in power point?

Or

(b) Explain the steps in printing Power Point presentations.

13. (a) Explain how tables are created, edited and formatted in Access database.

Or

(b) With an example database, explain the procedure to design and print reports in Access.

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CP–7823

Sub. Code
52

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircrafts Maintenance Science

AVIATION LEGISLATION – I

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define
   (a) Mandatory Modification
   (b) Airworthiness Directive.
   (c) Foreign airworthiness authority.
2. Write about aircraft log book.
3. Define identification plate.
4. Define registration fees.
5. Explain about issuance of certificate of airworthiness.
6. Define validity of certificate.
7. Define
   (a) Psychoactive Substances
   (b) Aircraft Goods
   (c) Accountable manager.
8. Write short note on manuals.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain the procedure for promulgation of CAR.

Or

(b) Explain documents relating to continued airworthiness of aircraft.

10. (a) Explain about the procedure for registration of aircraft.

Or

(b) Explain registration certificate and validity of registration of aircraft.

11. (a) Write about the procedure for documents technical literature.

Or

(b) Explain about the conditions for continued validity of C of A.

12. (a) Write the procedures for grant of approval to organizations.

Or

(b) What is quality system? Explain different types of manuals.

13. (a) Explain release as applicable to AO.

Or

(b) Explain the suspension or cancellation of certificate of airworthiness.

CP–7824

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

AVIATION MAINTENANCE PRACTICES

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write a note on torque wrenches.
2. What are objectives of servicing?
3. Write the classification of landing gear.
4. What are the significance of torque links?
5. Define Pre-ignition.
6. What is reversing Propeller?
7. What is propeller blade Shank?
8. Write short note on fretting corrosion.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) List out the special checks for flight through turbulent weather.

Or

(b) Name various maintenance manuals used in aviation and explain about them.

10. (a) Explain how the fixed landing gear of an aircraft is maintained.

Or

(b) Explain about the aircraft brake system maintenance procedure.

11. (a) Explain the steps for inspection of engine controls.

Or

(b) Describe the procedure of reciprocating engine shutdown.

12. (a) Describe the static balancing procedure of aircraft propeller.

Or

(b) Write notes on various forces acting on a propeller in flight.

13. (a) Describe in detail about metal spraying procedure.

Or

(b) Explain in detail about Granodizing process.

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CP–7825

Sub. Code
54

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

EXECUTIVE COMMUNICATION

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define medium.
2. What is Meta communication?
3. What is the purpose of business communication?
4. What is resolution?
5. Define formal report.
6. Explain tables and charts.
7. What is non-verbal communication?
8. What is Agenda?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain the principles of effective communication.

Or

(b) What are the essentials of business letter?

10. (a) What are the disadvantages of oral communication?

Or

(b) Why communication is significant for management?

11. (a) What are the different types of reports used for business? Explain.

Or

(b) Explain the structure of report.

12. (a) Explain the characteristics of Non-verbal communication.

Or

(b) Describe the details of body language.

13. (a) Write the procedure for conducting seminar.

Or

(b) Discuss about group discussion.

CP–7826

Sub. Code
61

B.Sc.(AMS) DEGREE EXAMINATION, NOVEMBER 2017

HELICOPTER CONFIGURATION AND MAINTENANCE

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write in brief about rigid rotor system.

2. Discuss in brief about "propeller region" on main rotor blade.

3. Write about "Telescopic method" of main rotor blade alignment.

4. Write about "Stick method" of tracking of main rotor blade.

5. Discuss in brief about "Stabilizer bar" fitted in Helicopter.

6. Write short note on "Blade sweeping".

7. Discuss in brief about dynamic balancing of Rotor blades.

8. (a) What are the factors that affect the rate of descent in autorotation?

   (b) Main rotor becomes more efficient, when the helicopter operates in ground effect. State the reason?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Discuss in detail about the constructional feature of solid main rotor mast.

Or

(b) Discuss in detail about the constructional feature hollow main rotor mast.

10. (a) What are the routine inspections carried out on main rotor?

Or

(b) State and discuss in detail about the special inspection carried out on main rotor.

11. (a) Write in detail about the features of metal main rotor blade.

Or

(b) Write in detail about the features of wooden main rotor blade.

12. (a) Discuss in detail about fully articulated rotor system.

Or

(b) Discuss in detail about semi rigid rotor system.

13. (a) Discuss in detail about autorotation in general.

Or

(b) Write in detail about "Ground effect" in helicopter.

CP–7827

Sub. Code
62

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

COMMUNICATION AND NAVIGATION SYSTEM

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Mention the functions of a radio transmitter.

2. Explain various types of microphones.

3. What is Aircraft SELCAL system? and explain in brief.

4. What is the importance of wave guide?

5. Explain about emergency locator transmitter.

6. Explain Antenna and its various types.

7. Describe about ATC transponder.

8. Name various radar bands with its frequency ranges.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) (i) Explain the aircraft VHF communication system.

(ii) Describe the Aircraft DME system.

Or

(b) (i) What is TCAS system and explain its function?

(ii) Describe Aircraft Automatic Direction Finder system.

10. (a) (i) Explain the various types of carrier wave.

(ii) List the various RF bands with its name frequency range and uses.

Or

(b) Explain ILS with neat sketch.

11. (a) (i) Write in brief Aircraft Radio Altimeter system.

(ii) Explain Aircraft Cockpit Voice Recorder.

Or

(b) Explain Global positioning system with its all segments.

12. (a) (i) Explain about aircraft INS system.

(ii) Write short notes on RMI.

Or

(b) Explain the aircraft VOR system with neat sketch.

13. (a) Write short notes on :

(i) Radar Safety.

(ii) Radom maintenance

Or

(b) Draw the block diagram of analog radar and explain its features.

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CP–7828

Sub. Code
63

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

AVIATION LEGISLATIONS — II

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define:
   (a) Damage Tolerant
   (b) Aerial Work
   (c) Commercial Operations.

2. Name the instruments that are fitted in gliders.

3. What are the instruments and equipments required for navigators?

4. Define
   (a) Light Aircraft
   (b) Heavy Aircraft
   (c) Concurrent Experience.

5. Explain:
   (a) Certificate of competency
   (b) Empty Weight
   (c) Scheduled Aircrafts.

6. What are the Documents carried on board on all flights?

7. Discuss in detail about "Emergency Locator Transmitters"(ELT).

8. Name different types of Flight Data Recorder (FDR).

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Write short notes on Flight Data Recorder (FDR).

Or

(b) What are the instruments and equipments required on aircraft operated at night?

10. (a) Explain the following terms.

(i) Approved Organization

(ii) Major Repair

(iii) Scheduled operator

(iv) Mandatory Modification.

Or

(b) Write down the in detail on the procedures for defect recording.

11. (a) Write in detail the number of First Aid Kit and Physician Kit as per the number of passenger's capacity of the aircraft.

Or

(b) Explain the procedure for getting special flight permit.

12. (a) Write down the items available in First Aid Kit and Universal Precaution Kit.

Or

(b) What are the Requirements and Procedures for issue of certificate of competency in welding and NDT?

13. (a) Discuss in detail about the operation of aircraft in IFR and VFR condition.

Or

(b) Briefly write the precautions and safety measures to be observed for safe loading of an aircraft.

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CP–7829

Sub. Code
64

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aircraft Maintenance Science

PRODUCTION AND MAINTENANCE MANAGEMENT

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What is the importance of production planning?
2. Define the term Total Quality management.
3. Why maintenance planning is important?
4. What are the types of product design?
5. List out any four steps to select a plant location.
6. What is the importance of reliability program?
7. What is meant by job scheduling?
8. Define the term work sampling.

Part B (5 × 12 = 60)

Answer **all** questions

9. (a) Explain the process of new product design in production management.

Or

(b) Explain the various function of production management.

10. (a) Describe the steps involved in aggregate planning.

Or

(b) Explain the advantages and limitations of PERT and CPM.

11. (a) Explain the barriers involved in Total Quality Management implementation.

Or

(b) What is meant by Time study? Explain the steps to conduct work sampling method.

12. (a) What is meant by reliability program? How it will help for maintenance planning?

Or

(b) Explain the various types of maintenance system.

13. (a) Explain the various features of inventory models.

Or

(b) Explain the roles and responsibilities of purchase manager in production department.

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